Aizaz Sharif

+92 305 9221199 :: Islamabad, 44000, Pakistan
aizazsharif@gmail.com :: https://aizazsharif.github.io
https://www.linkedin.com/in/AizazSharif

EDUCATION

Masters of Computer Science (MSCS) FAST-NUCES, Peshawar, Pakistan - GPA: 3.57/4.00 — August 2017-To Date
Data Analyst Nanodegree (DAND) Udacity — Dec 2016-August 2017
Deep Learning Nanodegree (DLND) Udacity — January 2017-July 2017
Bachelors of Computer Science (BSCS) FAST-NUCES, Peshawar, Pakistan - GPA: 3.84/4.00 — August 2012-May 2016

TECHNICAL SKILLS

Programming Languages: JAVA, C++/C, Python, JavaScript, Go, Bash
Machine Learning: Supervised Learning, Unsupervised Learning, Generative Models, Deep Learning, Natural Language Processing
Analytics: Jupyter, SQL, Tableau, Excel, R
Tools/Libraries: OpenCV, Tensorflow, Keras, Tesseract, NodeJs, Sklearn, Git, CUDA, Flask

KEY PROJECTS

Handwriting Generation Using Recurrent Neural Networks (GitHub) — June 2018
- PyTorch Implementation of Conditional + Unconditional Handwriting generation on 3-dimentional sequence dataset.
- Implemented Handwriting Synthesis with the help of LSTM and Attention Mechanism using Python.

Search and Sample Return (GitHub) — September 2017
- Utilizing Computer Vision to train Rover in a Mars Planet simulation capable of identifying rocks, obstacles and paths with high fidelity.
- Requires OpenCV in Python as well as Socket Programming.

Language Translation Using Deep Learning LSTM (GitHub) — July 2017
- Training a sequence-to-sequence model on a dataset of English and French sentences that can translate new sentences from English to French.
- Involves Tensorflow and Usage of LSTM Sequence-to-Sequence Model.

Generate Faces Using Deep Learning GANs (GitHub) — June 2017
- Using the latest innovation in Generative models to generate unseen data of faces.
- Uses Tensorflow to implement Convolutional Neural Network and Generative Adversarial Network.

WORK EXPERIENCE

National Center for Cyber Security (NCCS), Islamabad – Research Associate — February 2019-To Date
- Leading the team of software developers for the 'Mobile Phone Digital Forensics'.
- Working on a web application for the acquisition, analysis, and reporting of Android devices for criminal investigation.
- Involved in building backend microservices using Python and Flask.
- Utilizing HTML, CSS, and Javascript for the web application frontend.
- Using SQLite, SQLalchemy for dealing with device databases and Sphinx for automated code documentation.

FAST-NUCES, Peshawar - Research Assistant — September 2017-To Date
- Working under 'Colab' Research group for the campus.
- Involved in research areas related to GANs, Android Malware Detection, and Seismic Monitoring Network.
- Wrote and published a paper at 'Transactions on Emerging Telecommunications Technologies' journal.

DCUBE Technologies, Islamabad - Software Engineer (Machine Learning) — April 2017-June 2017
- Worked under the 'Product Innovation and Strategy' team.
- Integrated state-of-the-art OCR libraries including Google Vision, Omnipage, Ocropy and CLSTM to an ongoing C++ based large scale library.
- Tested Deep Learning based OCR libraries on letters images provided by Pitney Bowes and TCS.
- Increased the accuracy of the OCR prediction by 10% along with performance optimization.
- Gained Skills in OCR Pipeline, Python Linux Shell Scripting, Computer Vision and Deep Learning.

Techlogix, Islamabad - Software Engineer — October 2016-April 2017
- Worked as a Software Engineer for the Implementation and support of FLEXCUBE which is Oracle Universal Banking Solution.
- Implemented a workflow using PL/SQL for a smooth migration process from old to the proposed Oracle Solution under strict deadlines.
- Provided continuous support to the migrated system for running daily ongoing transactions with ease.

RESEARCH & PUBLICATIONS

- **Function Identification in Android Binaries Using Deep Learning** (In progress) — January 2019-To Date
- **Android Malware detection through Generative Adversarial Networks** (Published) — September 2017-May 2019

KEY ACHIEVEMENTS & AWARDS

- **7 Gold Medals and 1 Silver Medal** - FAST-NUCES, Peshawar, Pakistan — August 2012-May 2016
- **7 Deans list and 1 Rector List Certifications** - FAST-NUCES, Peshawar, Pakistan — August 2012-May 2016